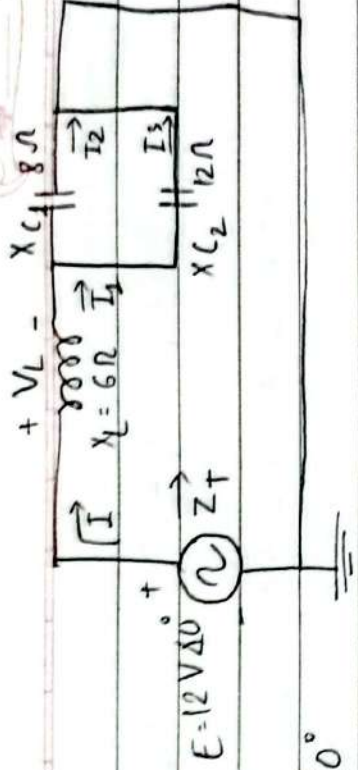


Tutorial-6

(1)



Soln:

$$Z_1 = j6 = 6 \angle 90^\circ$$

$$Z_2 = -j8 = 8 \angle -90^\circ$$

$$Z_3 = -j12 = 12 \angle -90^\circ$$

$$Z_T = Z_1 + \frac{Z_2 \cdot Z_3}{Z_2 + Z_3} = j6 + \frac{8 \angle -90^\circ \times 12 \angle -90^\circ}{-j8 - j12}$$

$$= j6 + \frac{96 \angle -180^\circ}{20 \angle -90^\circ}$$

$$= j6 + 4.8 \angle -90^\circ$$

$$= j6 - j4.8$$

$$= j1.2 = 1.2 \angle 90^\circ$$

$$I = \frac{E}{Z_T} = \frac{12 \angle 0^\circ}{1.2 \angle 90^\circ} = 10 A \angle -90^\circ$$

$$I_1 = I = 10 A \angle -90^\circ$$

$$I_2 = \frac{I_1 \times Z_3}{Z_2 + Z_3} = \frac{10 \angle -90^\circ \times 12 \angle -90^\circ}{-j8 - j12} = \frac{120 \angle -180^\circ}{20 \angle -90^\circ} = 6 A \angle -90^\circ$$

$$I_3 = I_1 - I_2 = 10 A \angle -90^\circ - 6 A \angle -90^\circ = 4 A \angle -90^\circ$$

$$V_L = Z_1 \times I_1 = 6 \angle 90^\circ \times 10 \angle -90^\circ = 60 V \angle 0^\circ$$

(2) Soln:

$$Z_1 = 560 + j560 = 791.959 \angle 45^\circ$$

$$Z_2 = 200 + \frac{-j400 \times -j400}{-j400 - j400} = 200 - j200 = 282.84 \angle -45^\circ$$

$$Z_T = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2} = \frac{791.959 \angle 45^\circ \times 282.84 \angle -45^\circ}{560 + j560 + 200 - j200}$$

$$= \frac{223997.6836 \angle 0^\circ}{840.951 \angle 25.346^\circ} = 266.362 \angle -25.346^\circ$$

$$\therefore I = \frac{100 \angle 0^\circ}{266.362 \angle -25.346^\circ} = 0.375 \angle 25.346^\circ$$

$$I_2 = \frac{0.375 \angle 25.346^\circ \times 791.959 \angle 45^\circ}{840.951 \angle 25.346^\circ} = 0.35315 \angle 45^\circ$$

$$\therefore V_c = 200 \angle -90^\circ \times 0.35315 \angle 45^\circ = 70.63 \angle -45^\circ$$

$$P = VI \cos \theta = 100 \times 0.375 \cos(25.346^\circ) = 33.89 \text{ W}$$

(3) Soln:

$$Z_1 = 2 - j2 = 2.828 \angle -45^\circ$$

$$Z_2 = 3 + j6 - j9 = 3 - j3 = 4.2426 \angle -45^\circ$$

$$Z_3 = 10$$

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} = 0.3536 \angle 45^\circ + 0.2357 \angle 45^\circ + 0.1 \angle 0^\circ$$

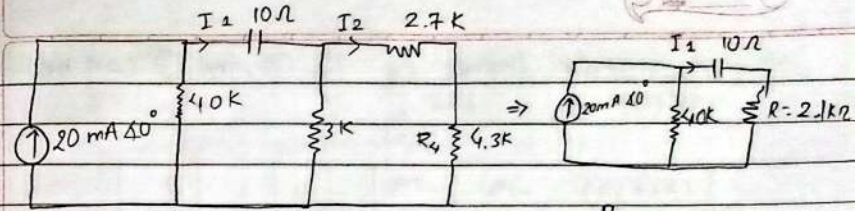
$$= 0.25 + j0.25 + 0.1666 + j0.1666 + 0.1$$

$$= 0.51666 + j0.41666$$

$$\text{or, } \frac{1}{Z} = 0.6637 \angle 38.884^\circ$$

$$\therefore I = \frac{E}{Z} = \frac{50 \angle 0^\circ}{0.6637 \angle 38.884^\circ} = 33.185 \text{ A} \angle -38.88^\circ$$

(4)



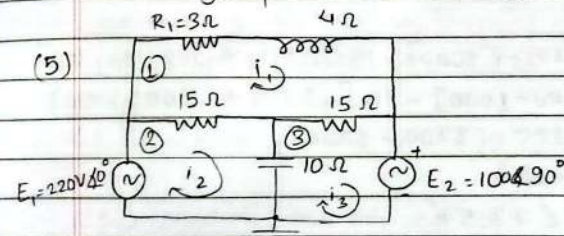
$$\therefore I_1 = \frac{20 \times 10^{-3} \times 40 \times 10^3}{40 \times 10^3 + 2.1 \times 10^3} = \frac{800}{42.1 \times 10^3} = 0.019 \angle -0.136^\circ$$

$$= \frac{800}{42100} \angle -0.136^\circ = 0.019 \angle -0.136^\circ$$

$$\therefore I_2 = \frac{I_1 \times 3 \times 10^3}{3 \text{ k} + 7 \text{ k}} = \frac{0.019 \angle -0.136^\circ \times 0.3}{0.3} = 5.7 \text{ mA} \angle -0.136^\circ$$

$$\therefore \text{Average power at } R_4 = I_2^2 \cdot R_4 = (5.7 \text{ mA})^2 \times 4.3 \text{ k}\Omega = 139.707 \text{ mW}$$

(5)



Applying KVL in loop 1,

$$(3 + j4)i_1 + 15(i_1 - i_2) + 15(i_1 - i_3) = 0$$

$$\text{or, } (33 + j4)i_1 - 15i_2 - 15i_3 = 0 \quad \text{--- (1)}$$

Applying KVL in loop 2,

$$220 = 15(i_2 - i_1) - j10(i_2 - i_3)$$

$$\text{or, } -15i_1 - (15 - j10)i_2 + j10i_3 = 220 \quad \text{--- (2)}$$

Applying KVL in loop 3,

$$15(i_3 - i_1) - j10(i_3 - i_2) = -j100$$

$$\text{or, } -15i_1 + j10i_2 + (15 - j10)i_3 = -j100 \quad \text{--- (3)}$$

∴ In matrix form eqⁿ ①, ②, and ③ can be written as,

$$\begin{bmatrix} (33+j4) & -15 & -15 \\ -15 & (15-j10) & j10 \\ -15 & j10 & (15-j10) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 220 \\ -j100 \end{bmatrix}$$

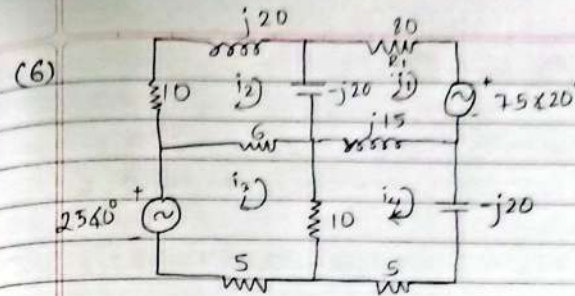
$$i_1 = \begin{bmatrix} 0 & -15 & -15 \\ 220 & (15-j10) & j10 \\ -j100 & j10 & (15-j10) \end{bmatrix} = \frac{A}{B}$$

$$\begin{bmatrix} (33+j4) & -15 & -15 \\ -15 & (15-j10) & j10 \\ -15 & j10 & (15-j10) \end{bmatrix}$$

$$\begin{aligned} A &= 15 [220(15-j10) + j^2 1000] - 15 [220 \times j10 + j100(15-j10)] \\ &= 15 [3300 - j2200 - 1000] - 15 [j2200 + j1500 + 1000] \\ &= 15 [2300 - j2200 - j3700 - 1000] \\ &= 15 [1300 - j5900] \\ &= 15 \times 6041.52 \angle -77.57^\circ \end{aligned}$$

$$\begin{aligned} B &= (33+j4) [(15-j10)^2 - (j10)^2] + 15 [-15(15-j10) + j50] - 15 [j150 + 15(15-j10)] \\ &= (33+j4)(225 - j300 - 100 + 100) + 15(-225 + j150 + j150) - 15(-j150 + 225 - j150) \\ &= (33+j4)(225 - j300) + 15(-225 + j300) - 15(225 - j300) \\ &= 15 [(33+j4)(15-j20) + (-225 + j300) - (225 - j300)] \\ &= 15 [495 - j600 + 80 - 225 + j300 - 225 + j300] \\ &= 15 \times 125 \end{aligned}$$

$$i_1 = \frac{A}{B} = \frac{15 \times 6041.52 \angle -77.57^\circ}{15 \times 125} = 48.33 \text{ A } \angle -77.57^\circ$$



The KVL eqⁿ for mesh 1 is,

$$\begin{aligned} 80i_1 + j15(i_1 - i_4) - j20(i_1 - i_2) - 75\angle 20^\circ \\ \text{or, } (80 - j5)i_1 - j15i_4 + j20i_2 = -70.477 - j25.6 \quad \text{--- ①} \end{aligned}$$

For mesh 2 the KVL equation is,

$$\begin{aligned} (10 + j20)i_2 - j20(i_2 - i_1) + 6(i_2 - i_3) = 0 \\ \text{or, } +j20i_1 + 16i_2 - 6i_3 = 0 \quad \text{--- ②} \end{aligned}$$

For mesh 3 the KVL eqⁿ is,

$$\begin{aligned} 5i_3 + 6(i_3 - i_2) + 10(i_3 - i_4) = 25 \\ -6i_2 + 21i_3 - 10i_4 = 25 \\ i_4 = -0.6i_2 + 2.1i_3 - 2.5 \quad \text{--- ③} \end{aligned}$$

For mesh 4, the KVL eqⁿ is

$$\begin{aligned} j15(i_4 - i_2) + 10(i_4 - i_3) + (5 - j20)i_4 = 0 \\ -j15i_2 - 10i_3 + (15 - j5)i_4 = 0 \quad \text{--- ④} \end{aligned}$$

From eqⁿ ③ and ④

$$\begin{aligned} -j15i_2 - 10i_3 + (15 - j5)(-0.6i_2 + 2.1i_3 - 2.5) = 0 \\ -j15i_2 - 10i_3 - (9 - 3j)i_2 + (31.5 - j10.5)i_3 - (37.5 - j12.5) = 0 \\ -j15i_2 - (9 - 3j)i_2 + (21.5 - j10.5)i_3 = 37.5 - j12.5 \quad \text{--- ⑤} \end{aligned}$$

Also from eqⁿ ① and ③

$$(80-j5)i_1 + j20i_2 - j15(-0.6i_2 + 2.1i_3 - 2.5) = -70.477 - j25.6$$

$$\text{or, } (80-j5)i_1 + j29i_2 - j31.5i_3 = -70.477 - j63.1 \quad \text{--- ⑥}$$

In matrix form eqⁿs ④, ⑤ & ⑥,

$$\begin{bmatrix} j20 & 16 & -6 \\ -j15 & -9+3j & 21.5-j10.5 \\ 80-j5 & j29 & -j31.5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 37.5-j12.5 \\ -70.477-j63.1 \end{bmatrix}$$

Using cramer's rule,

$$i_1 = \frac{\begin{vmatrix} 0 & 16 & -6 \\ 37.5-j12.5 & -9+3j & 21.5-j10.5 \\ -70.477-j63.1 & j29 & -j31.5 \end{vmatrix}}{\begin{vmatrix} j20 & 16 & -6 \\ -j15 & -9+3j & 21.5-j10.5 \\ 80-j5 & j29 & -j31.5 \end{vmatrix}} = \frac{A}{B}$$

$$A = -16 \{ (-37.5-j12.5)j31.5 - (-70.477-j63.1)(21.5-j10.5) \}$$

$$- 6 [j29(37.5-j12.5) - (-9+3j) \times (-70.477-j63.1)]$$

$$= -16 [-j1181.25 - 393.75 + 1515.255 + j1356.65 - j740 + 662.55]$$

$$- 6 [j1087.5 + 362.5 - 634.28 - j567.9 + j211.431 - 189.3]$$

$$= -16 [1784.055 - j564.6] - 6 [-461.09 + j731.03] = -25778.34 + j4647.42$$

$$= 26193.9177 \angle 169.78^\circ$$

$$B = j29 [(-9+3j)(-j31.5) - j29(21.5-j10.5)] - 16 [(-j15)(-j31.5) - (80-j5)(21.5-j10.5)] - 6 [-j15 \times j29 - (80-j5)(-9+3j)]$$

$$B = j20 [j283.5 + 94.5 - j623.5 - 304.5] - 16 [451.5 - 1720 + j840 + j107.5 + 52.5] - 6 [435 + 720 - j240 - j45 - 15]$$

$$= j20 [-210 - j340] - 16 [-2140 + j947.5] - 6 [1140 - j285]$$

$$= -j4200 + 6800 + 34240 - j25160 - 6840 + j1710$$

$$B = 34200 - j17650 = 38485.87 \angle -27.297^\circ$$

$$\therefore i_1 = \frac{A}{B} = \frac{26193.9177 \angle 169.78^\circ}{38485.87 \angle -27.29^\circ} = 0.68 \text{ A} \angle 197.05^\circ$$

(7) Soln:

$$\frac{V_1 - 50 \angle 120^\circ}{10} + \frac{V_1}{10} + \frac{V_1 - V_2}{4j} + \frac{V_1 - V_2}{-8j} = 0$$

$$\text{or, } 4j [V_1 + (25 - j43.3)] + 4j V_1 + 10(V_1 - V_2) - 5(V_1 - V_2) =$$

$$\text{or, } (5+8j)V_1 - 5V_2 = -173.2 - j100 \quad \text{--- ①}$$

At node 2,

$$\frac{V_2 - V_1}{-8j} + \frac{V_2 - V_1}{4j} + \frac{V_2}{2} + 0.8 \angle 70^\circ = 0$$

$$\text{or, } V_2 - V_1 + 4j V_2 + 2.1888j - 6.0136 = 0$$

$$\text{or, } -V_1 + (1+4j)V_2 = -j2.1888 + 6.0136 \quad \text{--- ②}$$

In matrix form, eqⁿ ① & ② can be written as

$$\begin{bmatrix} 5+8j & -5 \\ -1 & 1+4j \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -173.2 - j100 \\ 6.0136 - j2.188 \end{bmatrix}$$

using cramer's rule:

$$V_1 = \frac{\begin{vmatrix} -173.2 - j100 & -5 \\ 6.0136 - j2.188 & 1+4j \end{vmatrix}}{\begin{vmatrix} 5+8j & -5 \\ -1 & 1+4j \end{vmatrix}}$$

$$= \frac{-173.2 - j100 - j692.8 + 400 + 30.068 - j10.94}{5 + 20j + 8j - 32 - 5}$$

$$= \frac{843.788 \angle 72.276^\circ}{42.52 \angle 138.814^\circ}$$

$$= 19.844 \angle 211.09^\circ$$

$$V_2 = \frac{\begin{vmatrix} 5+8j & -173.2 - j100 \\ -1 & 6.0136 - j2.188 \end{vmatrix}}{\begin{vmatrix} 5+8j & -5 \\ -1 & 1+4j \end{vmatrix}}$$

$$= \frac{30.068 - j10.94 + j48.108 + 17.504 - 173.2 - j100}{5 + 20j + 8j - 32 - 5}$$

$$= \frac{140.464 \angle -153.428^\circ}{42.52 \angle 138.814^\circ}$$

$$= 3.3 \angle -292.238^\circ$$

(8) At node 1,

$$\frac{V_1 - 10}{1} + \frac{V_1 - V_2}{4} + \frac{V_1 - V_3}{4} = 0$$

$$\text{or, } 6V_1 - V_2 - V_3 = 40 \quad \text{--- (1)}$$

At node 2,

$$\frac{V_2 - V_3}{4} + \frac{V_2 - V_1}{4} + \frac{V_2}{j4} = 0$$

$$\text{or, } -jV_1 + (1+2j)V_2 - jV_3 = 0 \quad \text{--- (2)}$$

At node 3,

$$\frac{V_3 - V_2}{4} + \frac{V_3 - V_1}{4} + \frac{V_3}{-j4} = 0$$

$$\text{or, } jV_1 + jV_2 + (1-2j)V_3 = 0 \quad \text{--- (3)}$$

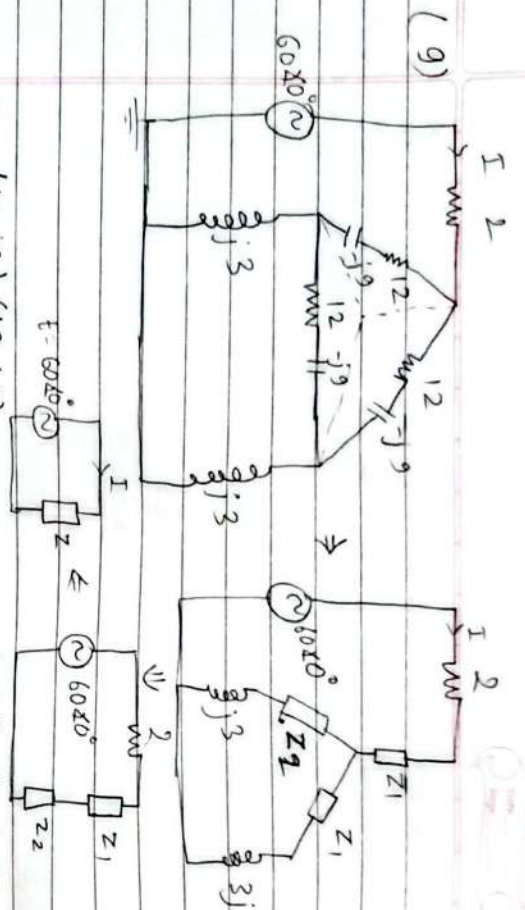
In matrix form, ①, ② & ③ can be written as,

$$\begin{bmatrix} 6 & -1 & -1 \\ -j & (1+2j) & -j \\ j & j & (1-2j) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \\ 0 \end{bmatrix}$$

using Cramer's rule,

$$V_3 = \frac{\begin{vmatrix} 6 & -1 & -1 \\ -j & (1+2j) & -j \\ j & j & (1-2j) \end{vmatrix}}{\begin{vmatrix} 6 & -1 & -1 \\ -j & 1+2j & -j \\ j & j & (1-2j) \end{vmatrix}} = \frac{40(1-j+2)}{6(1+4-1) + 1[j+2j^2+j^2]-1[-j^2-j(1+2j)]}$$

$$= \frac{120 - 40j}{24j - 3 + j - 3} = 7.027 \angle -18.435^\circ$$



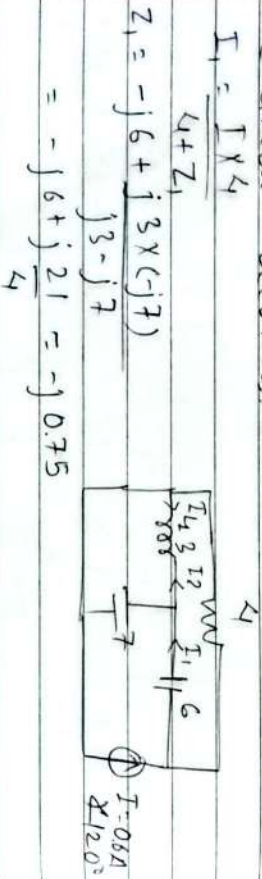
$$Z_1 = \frac{(12-j9)(12-j9)}{(12-j9) + (12-j9) + (12-j9)} = 4-j3 \Omega$$

$$Z_2 = \frac{(4-j3+3j)(4-j3+3j)}{(4-j3+3j) + (4-j3+3j)} = 2 \Omega$$

$$Z = 2 + 4-j3 + 2 = 8-j3 = 8.544 \angle -20.556^\circ$$

$$I = \frac{E}{Z} = \frac{60 \angle 0^\circ}{8.544 \angle -20.556^\circ} = 7.022 \text{ A } \angle 20.556^\circ$$

(10) considering the 120° source & killing the other source the circuit becomes.



$$I_1 = \frac{I \times 4}{4 + Z_1} = \frac{-j6 + j3 \times (-j7)}{j3-j7} = -j6 + j21 = -j0.75$$

$$I_1 = 0.6 \angle 120^\circ \times 4 = 0.5898 \angle 130.614^\circ$$

$$I_2 = \frac{-j7 \times I_1}{-j7 + j3} = \frac{4.1286 \angle 40.613^\circ}{4 \angle -90^\circ} = 1.03215 \angle 130.614^\circ$$

$$I_{L1} = -I_2 = -(1.03215 \angle 130.613^\circ)$$

considering 10V 490 source & killing other source, the circuit becomes,

$$Z_T = \frac{(4-j6) \times j3}{4-j6+j3} - j7$$

$$= \frac{21.633 \angle 33.69^\circ}{5 \angle -36.869^\circ} - j7$$

$$= 1.44 + j4.08 - j7 = 3.055 \angle -63.749^\circ$$

$$I = \frac{10 \angle 90^\circ}{3.255 \angle -63.749^\circ} = 3.072 \angle 153.749^\circ$$

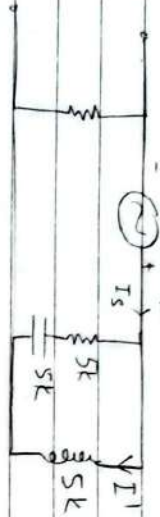
$$I_{L2} = \frac{I \times 4-j6}{j3+4-j6} = \frac{3.072 \angle 153.749^\circ \times 7.211 \angle -56.309^\circ}{5 \angle -36.869^\circ}$$

$$I_L = I_{L2} + I_{L1} = 4.4304 \angle 134.36^\circ - 1.03215 \angle 130.613^\circ$$

$$= -2.425627 + j2.38411$$

$$= 3.4 \angle 135.49^\circ$$

(11) Considering 20V source, and killing the other source the circuit becomes.



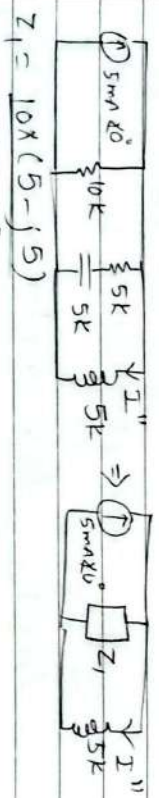
Here,

$$Z = \frac{(5-j5) \times j5 + 10}{5-j5+j5} = \frac{5+j5+10}{5+j5+10} = 15.811 \angle 18.435^\circ \text{ k}\Omega$$

$$I_3 = \frac{E}{Z} = \frac{1.2649 \angle -18.435^\circ \text{ mA}}{15.811 \angle 18.435^\circ \text{ k}\Omega}$$

$$I' = \frac{I_3 \times (5-j5)}{5-j5+5} = \frac{1.2649 \angle -18.435^\circ \times (1-j)}{5-j5+5} = 1.7888 \angle -63.435^\circ \text{ mA}$$

Considering 5 mA source and killing the other source the circuit becomes.

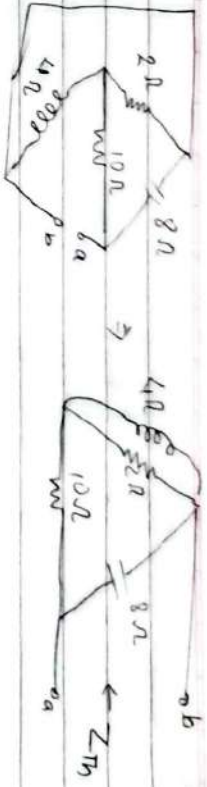


$$Z_1 = \frac{10 \times (5-j5)}{10+5-j5} = \frac{10-j10}{10+5-j5} = 4.472 \angle -26.565^\circ = 4-j2$$

$$I'' = \frac{Z_1 \times 5 \angle 0^\circ}{Z_1 + j5} = \frac{4.472 \angle -26.565^\circ \times 5}{4-j2+j5} = 4.472 \angle -63.435^\circ \text{ mA}$$

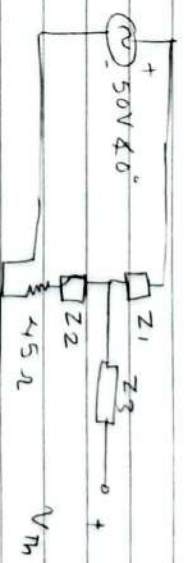
$$I = I' + I'' = 1.7888 \angle -18.435^\circ + 1.472 \angle -63.435^\circ = 6.261 \text{ mA} \angle -63.435^\circ$$

(12)



$$Z_{th} = \frac{\left[\frac{4 \times 2}{2+4} + 10 \right] \times (-j8)}{j4 \times 2 + 10 - j8} = \frac{-j160 + 384}{52 + j32} = 6.813 \angle -54.226^\circ$$

For V_{th} ,



$$Z_1 = \frac{2 \times (-8j)}{2-8j+10} = \frac{-16j}{12-8j} = 1.109 \angle -56.31^\circ \Omega$$

$$Z_2 = \frac{2 \times 10}{2-8j+10} = 1.3867 \angle 33.69^\circ = 1.1538 + j0.7692$$

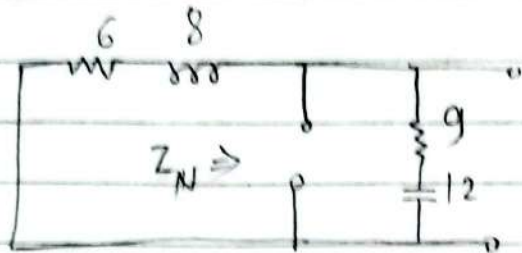
$$Z_3 = \frac{10 \times (-8j)}{2-8j+10} = \frac{-80j}{12-8j} = 5.547 \angle -56.31^\circ \Omega$$

$$V_{th} = 50 \times \left(\frac{Z_2 + j4\Omega}{Z_1 + Z_2 + j4\Omega} \right) = \frac{245.339 \angle 76.359^\circ}{4.2337 \angle 65.3026^\circ} = 57.949 \angle 11.056^\circ$$

The thevenin's equivalent circuit is

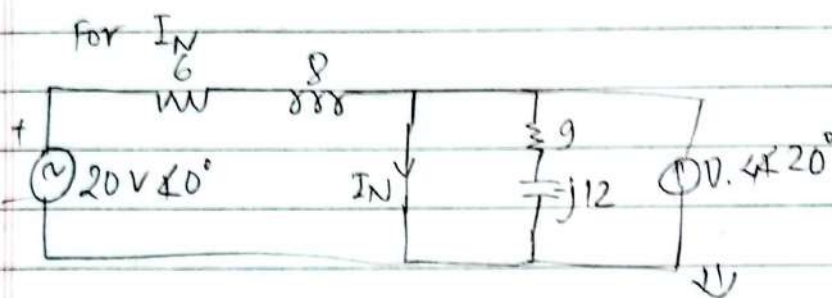


(13)

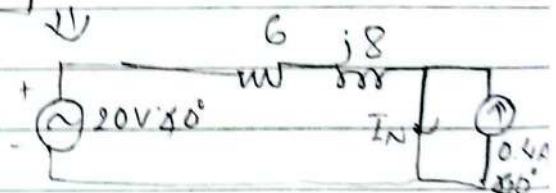


$$\begin{aligned} \therefore Z_N &= (9 - j12) (6 + j8) \\ &= \frac{9 - j12 + 6 + j8}{15 - j4} \\ &= \frac{54 - j72 + j72 + 96}{15 - j4} \end{aligned}$$

$$\therefore Z_N = \frac{150}{15.52 \angle -14.93^\circ} = 9.66 \angle 14.93^\circ$$



$$\begin{aligned} I_N &= \frac{20}{6 + j8} + 0.4 \angle 20^\circ \\ &= 2 \angle -53.13^\circ + 0.4 \angle 20^\circ \\ &= 2.15 \angle -42.87^\circ \end{aligned}$$



The Norton's equivalent circuit is

